

Supplementary Material for “Adaptive Partition-and-Combination Box for Power Flexibility Aggregation in Distribution Systems”

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I. COMPLETE ACTIVE DISTRIBUTION NETWORK MODEL

This supplementary material presents the complete polyhedral model of the distribution system (DS) used to construct the exact aggregation model (EAM). The DS is assumed to be radial and is represented by a directed graph $\mathcal{G}^{\text{DS}} = (\mathcal{N}, \mathcal{L})$, where bus 0 is the substation bus and every branch $(i, j) \in \mathcal{L}$ is oriented away from the substation. Let $\pi(i)$ denote the parent bus of bus i , and let $\mathcal{C}(i)$ denote the set of child buses of bus i . The set of operating periods is denoted by $\mathcal{T} = \{1, \dots, T\}$.

The sets of dispatchable generators (DGs), renewable energy sources (RESs), and energy storage systems (ESSs) connected to bus i are denoted by $\mathcal{G}_i$, $\mathcal{W}_i$, and $\mathcal{E}_i$, respectively. Empty summations are interpreted as zero. Positive substation active power indicates power imported from the upstream grid.

A. Linearized DistFlow Constraints

For each non-substation bus $i \in \mathcal{N} \setminus \{0\}$ and period $t \in \mathcal{T}$, the active- and reactive-power balance equations are

$$P_{\pi(i)i,t} = \sum_{j \in \mathcal{C}(i)} P_{ij,t} + P_{i,t}^D - \sum_{g \in \mathcal{G}_i} P_{g,t}^G - \sum_{w \in \mathcal{W}_i} P_{w,t}^W + \sum_{e \in \mathcal{E}_i} (P_{e,t}^{E,\text{ch}} - P_{e,t}^{E,\text{dis}}), \quad (\text{S1})$$

$$Q_{\pi(i)i,t} = \sum_{j \in \mathcal{C}(i)} Q_{ij,t} + Q_{i,t}^D - \sum_{g \in \mathcal{G}_i} Q_{g,t}^G - \sum_{w \in \mathcal{W}_i} Q_{w,t}^W - \sum_{e \in \mathcal{E}_i} Q_{e,t}^E. \quad (\text{S2})$$

Here, $P_{i,t}^D$ and $Q_{i,t}^D$ are the fixed active and reactive loads; $P_{ij,t}$ and $Q_{ij,t}$ are the branch power flows; and $P_{g,t}^G$, $P_{w,t}^W$, $P_{e,t}^{E,\text{ch}}$, and $P_{e,t}^{E,\text{dis}}$ denote the DG output, RES output, ESS charging power, and ESS discharging power, respectively.

Let $v_{i,t}$ denote the squared voltage magnitude at bus i . The linearized voltage-drop equation is

$$v_{j,t} = v_{i,t} - 2(r_{ij}P_{ij,t} + x_{ij}Q_{ij,t}), \quad (i, j) \in \mathcal{L}, \quad t \in \mathcal{T}, \quad (\text{S3})$$

where r_{ij} and x_{ij} are the resistance and reactance of branch (i, j) . The substation voltage and nodal voltage limits are

$$v_{0,t} = (V_0^{\text{ref}})^2, \quad t \in \mathcal{T}, \quad (\text{S4})$$

$$(\underline{V}_i)^2 \leq v_{i,t} \leq (\bar{V}_i)^2, \quad i \in \mathcal{N}, \quad t \in \mathcal{T}. \quad (\text{S5})$$

To preserve linearity, the branch apparent-power limit is represented by an inscribed regular polygon. Let $\mathcal{M} = \{1, \dots, M\}$ and $\theta_m = 2\pi(m-1)/M$. Then

$$P_{ij,t} \cos \theta_m + Q_{ij,t} \sin \theta_m \leq \bar{S}_{ij} \cos \left(\frac{\pi}{M} \right), \quad (i, j) \in \mathcal{L}, \quad t \in \mathcal{T}, \quad m \in \mathcal{M}, \quad (\text{S6})$$

where $\bar{S}_{ij}$ is the branch apparent-power rating. Increasing M improves the accuracy of the polygonal approximation while retaining a linear formulation.

B. Dispatchable Generator Constraints

For each DG $g \in \mathcal{G}_i$ and period $t \in \mathcal{T}$,

$$\underline{P}_g^G \leq P_{g,t}^G \leq \bar{P}_g^G, \quad (\text{S7})$$

$$\underline{Q}_g^G \leq Q_{g,t}^G \leq \bar{Q}_g^G. \quad (\text{S8})$$

The intertemporal ramping limits are

$$-R_g^{G,\text{dn}} \leq P_{g,t}^G - P_{g,t-1}^G \leq R_g^{G,\text{up}}, \quad g \in \mathcal{G}_i, \quad t = 2, \dots, T. \quad (\text{S9})$$

C. Renewable Energy Source Constraints

For each RES $w \in \mathcal{W}_i$ and period $t \in \mathcal{T}$, the utilized renewable power is bounded by its forecasted available power:

$$0 \leq P_{w,t}^W \leq \hat{P}_{w,t}^W, \quad w \in \mathcal{W}_i, \quad t \in \mathcal{T}. \quad (\text{S10})$$

Accordingly, the curtailed renewable power is

$$P_{w,t}^{W,\text{cur}} = \hat{P}_{w,t}^W - P_{w,t}^W, \quad P_{w,t}^{W,\text{cur}} \geq 0. \quad (\text{S11})$$

The inverter capability of the RES is represented by

$$P_{w,t}^W \cos \theta_m + Q_{w,t}^W \sin \theta_m \leq \bar{S}_w^W \cos \left(\frac{\pi}{M} \right), \quad w \in \mathcal{W}_i, \quad t \in \mathcal{T}, \quad m \in \mathcal{M}. \quad (\text{S12})$$

If reactive-power regulation by RESs is not considered, (S12) can be replaced by $Q_{w,t}^W = 0$ or by a prescribed power-factor relation.

D. Energy Storage System Constraints

For each ESS $e \in \mathcal{E}_i$ and period $t \in \mathcal{T}$, the charging and discharging powers satisfy

$$0 \leq P_{e,t}^{E,\text{ch}} \leq \bar{P}_e^{E,\text{ch}}, \quad (\text{S13})$$

$$0 \leq P_{e,t}^{E,\text{dis}} \leq \bar{P}_e^{E,\text{dis}}. \quad (\text{S14})$$

To retain a convex polyhedral model without introducing binary operating-mode variables, the convex-hull representation of the charging/discharging disjunction is adopted:

$$\frac{P_{e,t}^{E,\text{ch}}}{\bar{P}_e^{E,\text{ch}}} + \frac{P_{e,t}^{E,\text{dis}}}{\bar{P}_e^{E,\text{dis}}} \leq 1, \quad e \in \mathcal{E}_i, \quad t \in \mathcal{T}. \quad (\text{S15})$$

The energy dynamics are

$$E_{e,t} = E_{e,t-1} + \eta_e^{\text{ch}} \Delta t P_{e,t}^{E,\text{ch}} - \frac{\Delta t}{\eta_e^{\text{dis}}} P_{e,t}^{E,\text{dis}}, \quad e \in \mathcal{E}_i, \quad t \in \mathcal{T}, \quad (\text{S16})$$

with

$$\underline{E}_e \leq E_{e,t} \leq \bar{E}_e, \quad e \in \mathcal{E}_i, \quad t \in \mathcal{T}, \quad (\text{S17})$$

$$E_{e,0} = E_e^{\text{ini}}, \quad e \in \mathcal{E}_i, \quad (\text{S18})$$

$$E_{e,T} = E_e^{\text{tar}}, \quad e \in \mathcal{E}_i. \quad (\text{S19})$$

A cyclic scheduling condition is obtained by setting $E_e^{\text{tar}} = E_e^{\text{ini}}$.

Define the net ESS active-power injection as

$$P_{e,t}^E = P_{e,t}^{E,\text{dis}} - P_{e,t}^{E,\text{ch}}. \quad (\text{S20})$$

The ESS inverter capability is then represented by

$$P_{e,t}^E \cos \theta_m + Q_{e,t}^E \sin \theta_m \leq \bar{S}_e^E \cos \left(\frac{\pi}{M} \right), \quad e \in \mathcal{E}_i, \quad t \in \mathcal{T}, \quad m \in \mathcal{M}. \quad (\text{S21})$$

E. Substation Power and Projection Variable

The active and reactive powers exchanged with the upstream grid are

$$P_t^{\text{sub}} = \sum_{j \in \mathcal{C}(0)} P_{0j,t}, \quad t \in \mathcal{T}, \quad (\text{S22})$$

$$Q_t^{\text{sub}} = \sum_{j \in \mathcal{C}(0)} Q_{0j,t}, \quad t \in \mathcal{T}. \quad (\text{S23})$$

The substation transformer limit can be represented by

$$P_t^{\text{sub}} \cos \theta_m + Q_t^{\text{sub}} \sin \theta_m \leq \bar{S}^{\text{sub}} \cos \left(\frac{\pi}{M} \right), \quad t \in \mathcal{T}, \quad m \in \mathcal{M}. \quad (\text{S24})$$

The projected variable is the multi-period substation active-power trajectory

$$\mathbf{p} = \text{col} (P_1^{\text{sub}}, P_2^{\text{sub}}, \dots, P_T^{\text{sub}}). \quad (\text{S25})$$

Let $\mathbf{y}$ collect the controllable DER variables and substation power variables, and let $\mathbf{z}$ collect the branch-flow, voltage, and energy-state variables. Constraints (S1)–(S24) can be compactly expressed as

$$\mathbf{F}\mathbf{y} + \mathbf{E}\mathbf{z} \leq \mathbf{f}, \quad \mathbf{p} = \mathbf{C}\mathbf{y}. \quad (\text{S26})$$

Therefore, the EAM is the projection of the DS feasible region onto the substation active-power trajectory:

$$\Omega = \{\mathbf{p} \mid \exists \mathbf{y}, \mathbf{z} : \mathbf{F}\mathbf{y} + \mathbf{E}\mathbf{z} \leq \mathbf{f}, \mathbf{p} = \mathbf{C}\mathbf{y}\}. \quad (\text{S27})$$

Because all constraints in (S1)–(S24) are linear and no binary variables are introduced, the DS feasible region and its projection Ω are convex polyhedra.